1 Boson case

1.1 Imaginary-time

The influence functional is

$$I[s(\tau)] = e^{-\int_0^\beta d\tau \int_0^\beta d\tau' s(\tau)\Delta(\tau,\tau')s(\tau')}, \tag{1}$$

where

$$\Delta(\tau,\tau') = \int d\omega J(\omega) D_\omega(\tau,\tau'). \tag{2}$$

Here D_ω is the free bath Green's function that

$$D_\omega = \begin{cases} -[1+n(\omega)]e^{-\omega(\tau-\tau')} = -(1-e^{-\beta\omega})^{-1}e^{-\omega(\tau-\tau')}, & \tau \geqslant \tau' \\ -n(\omega)e^{-\omega(\tau-\tau')} = -(e^{\beta\omega}-1)^{-1}e^{-\omega(\tau-\tau')}, & \tau < \tau' \end{cases}. \tag{3}$$

The QUAPI discretization is ($e^{\sum_{jk} s_j \Delta_{jk} s_k}$)

$$\Delta_{jk} = \int_{j\Delta t}^{(j+1)\Delta t} d\tau \int_{k\Delta t}^{(k+1)\Delta t} d\tau' \Delta(\tau,\tau'), \tag{4}$$

that is

$$\Delta_{jk} = \begin{cases} -2\int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}e^{-\omega(j-k)\Delta\tau}(1-\cosh\omega\Delta\tau), & j>k \\ -\int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}[e^{-\omega\Delta\tau}-(1-\omega\Delta\tau)] - \int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}[e^{\omega\Delta\tau}-(1+\omega\Delta\tau)], & j=k \\ -2\int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}e^{-\omega(j-k)\Delta\tau}(1-\cosh\omega\Delta\tau), & j<k \end{cases}. \tag{5}$$

When $\omega \to 0$, we have

$$(1-e^{-\beta\omega})^{-1} = (\beta\omega)^{-1}, \quad (e^{\beta\omega}-1)^{-1} = (\beta\omega)^{-1}, \quad (1-\cosh\omega\Delta\tau) = \frac{\omega^2\Delta\tau^2}{2}, \quad [e^{-\omega\Delta\tau}-(1-\omega\Delta\tau)] = [e^{\omega\Delta\tau}-(1+\omega\Delta\tau)] = \frac{\omega^2\Delta\tau^2}{2} \tag{6}$$

Thus

$$\Delta_{jk} = \begin{cases} -\frac{\Delta\tau^2}{\beta}\int d\omega \frac{J(\omega)}{\omega} e^{-\omega(j-k)\Delta\tau} & j>k \\ -\frac{\Delta\tau^2}{\beta}\int d\omega \frac{J(\omega)}{\omega} & j=k \\ -\frac{\Delta\tau^2}{\beta}\int d\omega \frac{J(\omega)}{\omega} e^{-\omega(j-k)\Delta\tau} & j<k \end{cases} \tag{7}$$

The integral $\int_0^\varepsilon x^\alpha dx$ converges if $\alpha > -1$.

1.2 Keldysh contour

The influence functional is

$$I[s(t)] = e^{-\int_C dt \int_C dt' s(t)\Delta(t,t')s(t')}, \tag{8}$$

where

$$\Delta(t,t') = i\int d\omega J(\omega) D_\omega(t,t'). \tag{9}$$

Here $D_\omega(t,t')$ is the free bath Green's function that

$$D_\omega(t,t') = -i\langle T_C b_\omega(t) b_\omega^\dagger(t')\rangle. \tag{10}$$

The QUAPI discretization is

$$\Delta_{jk}^{++} = \begin{cases} 2\int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t), & j>k \\ \int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}[(1-i\omega\Delta t)-e^{-i\omega\Delta t}] + \int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}[(1+i\omega\Delta t)-e^{i\omega\Delta t}], & j=k \\ 2\int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t), & j<k \end{cases} \tag{11}$$

$$\Delta_{jk}^{--} = \begin{cases} 2\int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) & j>k \\ \int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}[(1-i\omega\Delta t)-e^{-i\omega\Delta t}] + \int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}[(1+i\omega\Delta t)-e^{i\omega\Delta t}] & j=k \\ 2\int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) & j<k \end{cases} \tag{12}$$

$$\Delta_{jk}^{+-} = 2\int d\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) \tag{13}$$

$$\Delta_{j,k}^{-+} = 2\int d\omega \frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) \tag{14}$$

When $\omega \to 0$, we have

$$(1-e^{-\beta\omega})^{-1} = (\beta\omega)^{-1}, \quad (e^{\beta\omega}-1)^{-1} = (\beta\omega)^{-1}, \quad 1-\cos\omega\Delta t = \frac{\omega^2\Delta t^2}{2}, \quad e^{-i\omega\Delta t} = 1-i\omega\Delta t - \frac{\omega^2\Delta t^2}{2}, \quad e^{i\omega\Delta t} = 1+i\omega\Delta t - \frac{\omega^2\Delta t^2}{2}, \tag{15}$$

thus

$$\Delta_{jk}^{++}=\begin{cases}\frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}e^{-i\omega(j-k)\Delta t} & j>k\\ \frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2} & j=k\\ \frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}e^{-i\omega(j-k)\Delta t} & j<k\end{cases} \tag{16}$$

$$\Delta_{jk}^{--}=\begin{cases}\frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega}e^{-i\omega(j-k)\Delta t} & j>k\\ \frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega} & j=k\\ \frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega}e^{-i\omega(j-k)\Delta t} & j>k\end{cases} \tag{17}$$

$$\Delta_{jk}^{+-}=\Delta_{jk}^{-+}=\frac{\Delta t^2}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega}e^{-i\omega(j-k)\Delta t} \tag{18}$$

1.3 Kadanoff contour

The extra QUAPI discretization terms are

$$\Delta_{jk}^{13}=\Delta_{jk}^{23}=\int_{j\Delta t}^{(j+1)\Delta t}\mathrm{d}t\int_{k\Delta\tau}^{(k+1)\Delta\tau}\int \mathrm{d}\omega J(\omega)(e^{\beta\omega}-1)^{-1}e^{-i\omega t}e^{\omega\tau}, \tag{19}$$

$$\Delta_{jk}^{31}=\Delta_{jk}^{32}=\int_{j\Delta t}^{(j+1)\Delta t}\mathrm{d}t\int_{k\Delta\tau}^{(k+1)\Delta\tau}\int \mathrm{d}\omega J(\omega)(1-e^{-\beta\omega})^{-1}e^{-\omega\tau}e^{i\omega t}. \tag{20}$$

that is

$$\Delta_{jk}^{13}=\Delta_{jk}^{23}=i\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(e^{\beta\omega}-1)^{-1}e^{-i\omega j\Delta t}e^{\omega k\Delta\tau}(1-e^{-i\omega\Delta t})(1-e^{\omega\Delta\tau}), \tag{21}$$

$$\Delta_{jk}^{31}=\Delta_{jk}^{32}=i\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(1-e^{-\beta\omega})^{-1}e^{-\omega j\Delta\tau}e^{i\omega k\Delta t}(1-e^{i\omega\Delta t})(1-e^{-\omega\Delta\tau}). \tag{22}$$

When $\omega\to 0$, we have

$$(e^{\beta\omega}-1)^{-1}=(\beta\omega)^{-1},\quad (1-e^{-\beta\omega})^{-1}=(\beta\omega)^{-1},\quad (1-e^{-i\omega\Delta t})=i\omega\Delta t,\quad (1-e^{\omega\Delta\tau})=-\omega\Delta\tau,\quad (1-e^{i\omega\Delta\tau})=-i\omega\Delta\tau,\quad (1-e^{-\omega\Delta\tau})=\omega\Delta\tau, \tag{23}$$

thus

$$\Delta_{jk}^{13}=\Delta_{jk}^{23}=\frac{\Delta\tau\Delta t}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega}e^{-i\omega j\Delta t}e^{\omega k\Delta\tau},\quad \Delta_{jk}^{31}=\Delta_{jk}^{32}=\frac{\Delta\tau\Delta t}{\beta}\int \mathrm{d}\omega\frac{J(\omega)}{\omega}e^{-\omega j\Delta\tau}e^{i\omega k\Delta t}. \tag{24}$$

2 Fermion Case

2.1 Imaginary-time

The influence functional is

$$I[\bar{a}a]=e^{-\int_0^\beta \mathrm{d}\tau\int_0^\beta \mathrm{d}\tau' s(\tau)\Delta(\tau,\tau')s(\tau')},\quad \Delta(\tau,\tau')=\int J(\omega)D_\omega(\tau,\tau'), \tag{25}$$

where

$$D_\omega(\tau,\tau')=-\begin{cases}-[1-f(\omega)]e^{-\omega(\tau-\tau')}=-(1+e^{-\beta\omega})e^{-\omega(\tau-\tau')} & \tau\geqslant\tau'\\ f(\omega)e^{-\omega(\tau-\tau')}=(e^{\beta\omega}+1)^{-1}e^{-\omega(\tau-\tau')} & \tau<\tau'\end{cases} \tag{26}$$

The QUAPI discretization is

$$\Delta_{jk}=\begin{cases}2\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1}e^{-\omega(j-k)\Delta\tau}(1-\cosh\omega\Delta\tau) & j>k\\ \int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1}[e^{-\omega\Delta\tau}-(1-\omega\Delta\tau)]-\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1}[e^{\omega\Delta\tau}-(1+\omega\Delta\tau)] & j=k\\ -2\int \mathrm{d}\omega\frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1}e^{-\omega(j-k)\Delta\tau}(1-\cosh\omega\Delta\tau) & j<k\end{cases} \tag{27}$$

When $\omega\to 0$, we have

$$(1-\cosh\omega\Delta\tau)=\frac{\omega^2\Delta\tau^2}{2},\quad e^{-\omega\Delta\tau}-(1-\omega\Delta\tau)=\frac{\omega^2\Delta\tau^2}{2},\quad e^{\omega\Delta\tau}-(1+\omega\Delta\tau)=\frac{\omega^2\Delta\tau^2}{2}, \tag{28}$$

thus

$$\Delta_{jk}=\begin{cases}\Delta\tau^2\int \mathrm{d}\omega J(\omega)(1+e^{-\beta\omega})^{-1}e^{-\omega(j-k)\Delta\tau}, & j>k\\ \frac{\Delta\tau^2}{2}\int \mathrm{d}\omega J(\omega), & j=k\\ -\Delta\tau^2\int \mathrm{d}\omega J(\omega)(e^{\beta\omega}+1)^{-1}e^{-\omega(j-k)\Delta\tau}, & j<k\end{cases} \tag{29}$$

2.2 Keldysh contour

The influence functional is

$$I[\bar{a}a]=e^{-\int_C \mathrm{d}t\int_C \mathrm{d}t'\bar{a}(t)\Delta(t,t')a(t')}, \tag{30}$$

where

$$\Delta(t,t') = i\mathscr{P}_{tt'}\int \mathrm{d}\omega J(\omega) D_\omega(t,t'). \tag{31}$$

Here $D_\omega(t,t')$ is the free bath Green's function that

$$D_\omega(t,t') = -i\langle T_C c_\omega(t) c_\omega^\dagger(t')\rangle. \tag{32}$$

The QUAPI discretization is

$$\Delta_{jk}^{++} = \begin{cases} 2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t), & j>k \\ \int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1}[(1-i\omega\Delta t)-e^{-i\omega\Delta t}] - \int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1}[(1+i\omega\Delta t)-e^{i\omega\Delta t}], & j=k \\ -2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) & j<k \end{cases} \tag{33}$$

$$\Delta_{jk}^{--} = \begin{cases} -2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) & j>k \\ \int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1}[(1+i\omega\Delta t)-e^{i\omega\Delta t}] - \int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1}[(1-i\omega\Delta t)-e^{-i\omega\Delta t}] & j=k \\ 2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) & j<k \end{cases} \tag{34}$$

$$\Delta_{jk}^{+-} = 2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) \tag{35}$$

$$\Delta_{jk}^{-+} = -2\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t}(1-\cos\omega\Delta t) \tag{36}$$

When $\omega\to 0$, we have

$$1-\cos\omega\Delta t = \frac{\omega^2\Delta t^2}{2}, \quad (1-i\omega\Delta t)-e^{-i\omega\Delta t} = \frac{\omega^2\Delta t^2}{2}, \quad (1+i\omega\Delta t)-e^{i\omega\Delta t} = \frac{\omega^2\Delta t^2}{2} \tag{37}$$

thus

$$\Delta_{jk}^{++} = \begin{cases} \Delta t^2 \int \mathrm{d}\omega J(\omega)(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t} & j>k \\ \frac{\Delta t^2}{2}\int \mathrm{d}\omega J(\omega) & j=k \\ -\Delta t^2 \int \mathrm{d}\omega J(\omega)(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t} & j<k \end{cases} \tag{38}$$

$$\Delta_{jk}^{--} = \begin{cases} -\Delta t^2 \int \mathrm{d}\omega J(\omega)(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t} & j>k \\ \frac{\Delta t^2}{2}\int \mathrm{d}\omega J(\omega) & j=k \\ \Delta t^2 \int \mathrm{d}\omega J(\omega)(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t} & j<k \end{cases} \tag{39}$$

$$\Delta_{jk}^{+-} = \Delta t^2 \int \mathrm{d}\omega J(\omega)(e^{\beta\omega}+1)^{-1} e^{-i\omega(j-k)\Delta t} \tag{40}$$

$$\Delta_{jk}^{-+} = -\Delta t^2 \int \mathrm{d}\omega J(\omega)(1+e^{-\beta\omega})^{-1} e^{-i\omega(j-k)\Delta t} \tag{41}$$

2.3 Kadanoff contour

On Kadanoff contour, the hybridization function contains no $\mathscr{P}_{tt'}$, thus Δ_{jk}^{+-} and Δ_{jk}^{-+} becomes $-\Delta_{jk}^{+-}$ and $-\Delta_{jk}^{-+}$ on Keldysh contour. The extra QUAPI terms are

$$\Delta_{jk}^{13} = \Delta_{jk}^{23} = -i\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(e^{\beta\omega}+1)^{-1} e^{-i\omega j\Delta t} e^{\omega k\Delta\tau}(1-e^{-i\omega\Delta t})(1-e^{\omega\Delta\tau}) \tag{42}$$

$$\Delta_{jk}^{31} = \Delta_{jk}^{32} = i\int \mathrm{d}\omega \frac{J(\omega)}{\omega^2}(1+e^{-\beta\omega})^{-1} e^{-\omega j\Delta\tau} e^{i\omega k\Delta t}(1-e^{i\omega\Delta t})(1-e^{-\omega\Delta\tau}) \tag{43}$$

When $\omega\to 0$, we have

$$1-e^{-i\omega\Delta t} = i\omega\Delta t, \quad 1-e^{\omega\Delta\tau} = -\omega\Delta\tau, \quad 1-e^{i\omega\Delta t} = -i\omega\Delta t, \quad 1-e^{-\omega\Delta\tau} = \omega\Delta\tau \tag{44}$$

thus

$$\Delta_{jk}^{13} = \Delta_{jk}^{23} = -\Delta t\Delta\tau \int \mathrm{d}\omega J(\omega)(e^{\beta\omega}+1)^{-1} e^{-i\omega j\Delta t} e^{\omega k\Delta\tau} \tag{45}$$

$$\Delta_{jk}^{31} = \Delta_{jk}^{32} = \Delta t\Delta\tau \int \mathrm{d}\omega J(\omega)(1+e^{-\beta\omega})^{-1} e^{-\omega j\Delta\tau} e^{i\omega k\Delta t} \tag{46}$$